Name: _____ Class: _____ Date: _____

Marathon Runner: Water Balance

Why did a runner who drank plenty of water end up in the Medical Tent? In this model you control how she runs and what she drinks, then watch what happens to the water and salt in her blood.

Open: sims.mr-miner.com/play/water-balance

How to use the model

1. Choose the Pace, How much the runner drinks, What the runner drinks and the Race length.
2. Press Start race (or Space). Use Speed 4x if you want it faster; the results are the same.
3. When the race is over, read the Blood sodium gauge and the observation sentence under the body.
4. To compare two runs, press Keep this run, change a setting, then press Run again. The kept run shows as a dashed line on the graph.

Normal blood sodium is 135–145 mEq/L (the green band). If the runner collapses, the race stops and the screen tells you why. Write that down: it is a result, not a mistake.

Part 1: Get to know the screen

Before you start a race, find each part of the body diagram. What does each one show?

Part of the screen	What it shows
The cup and its arrow	
The blood box: blue water level	
The blood box: yellow diamonds	
The purple ADH meter	
The kidneys and the urine arrow	

Part 2: Predict, then test

Use a 4-hour race and Pace: Run unless the table says otherwise. Before each run, predict whether her blood sodium will end up too high, normal, or too low. Then run it and record what happens.

Run	Settings	Prediction (high / normal / low)	Sodium at end (mEq/L)	ADH at end (high / med / low)	Urine (a lot / some / a little)	Collapsed? Why?
A	Run, None					
B	Run, Medium, Water					
C	Run, Lots, Water					
D	Run, Lots, Sports drink					
E	Stop, Lots, Water					
F	8 hours, Run, Lots, Water					

1. Which runs ended outside the normal range? Which direction was each one?

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2. In Run C, what did the observation sentence say about ADH and the kidneys? Why couldn't the kidneys bring her sodium back to normal?

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Part 3: Compare two runners

Run C again and press Keep this run. Then change Pace to Stop (Run E) and press Run again.

3. Which runner's blood was more diluted at the end, the one running or the one stopped? Use what sweat does to explain why.

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4. Run A (Run, None) and then Stop with None. Copy the Salt row of the "Water and salt so far" table.

	Salt lost in sweat	Sodium at end (mEq/L)
Run, None		
Stop, None		

Which runner lost more salt? Which one has saltier blood? How can both be true? (Hint: look at the Water row too.)

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5. Compare Run C (water) and Run D (sports drink). Did the sports drink prevent the problem? What does that tell you about the main cause?

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Part 4: Back to the Medical Tent

Use your printed Medical Tent Data.

	Blood sodium (mEq/L)	Normal range?
Our runner (from the Medical Tent Data)		
The model's closest run (which one?)		

6. Which settings in the model come closest to what happened to our runner? Describe how she probably ran and drank.

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7. When her blood became diluted, which way did water move: into her cells or out of them? (Check the observation sentence.) Why would that be dangerous for brain cells?

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Part 5: Explain it

8. Claim, evidence, reasoning: Why did a runner who drank lots of water end up with low blood sodium? Use at least two numbers from the model and the words ADH, kidneys and diluted.

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9. What advice would you give a runner about drinking during a long race? What does this model leave out that a real runner might also need to think about?

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Remember: this is a simplified model. Its numbers act like a real body, but they are not medical measurements.